

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 3 – Class Test

Course Code:	DSA0307	Course Title:	Natural Language Processing
CO Assessed:	CO2 – Manipulation	Assessment:	Assessment Tool 3 – Class Test
Weightage:	20% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
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Section / Batch:	CSE-AI	Date:	

Code of Conduct

I **A.Sai Rohit (192472144)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: **A.Sai Rohit**

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
N-gram Models and Word Counting for Language Modeling	N-gram Construction, Word Frequency Analysis, Probability Estimation	1
Unsmoothed N-grams and Smoothing Techniques in Syntax Analysis	Unsmoothed N-grams, Backoff, Deleted Interpolation	2
Entropy and Language Uncertainty in NLP Systems	Information Theory, Entropy Measurement, Language Prediction	3
English Word Classes and Tagsets for Linguistic Analysis	Word Categories, POS Categories, English Tagsets	4
Part-of-Speech Tagging Techniques and Applications	POS Tagging, Rule-Based Tagging, Stochastic Tagging, Transformation-Based Tagging	5

Annexure A – Class Test

1. A document processing system encounters the words “talked”, “talking”, and “talks”. Apply inflectional morphology rules and Porter Stemmer steps to derive the normalized root form. Identify the suffixes removed and explain the role of morphological normalization in improving information retrieval accuracy.
2. Given the input words: “relational”, “conditional”, “happiness”, “running”, design a hybrid morphological processing pipeline using derivational rules, inflectional analysis, finite-state morphological parsing, and Porter Stemmer. Decompose each word into root and affixes, represent transitions using finite-state parsing logic, and apply stemming steps to obtain the final normalized forms. Analyze how each technique contributes to accurate root extraction.
3. An enterprise-level search engine optimization (SEO) system processes web content containing variations such as “optimize”, “optimization”, “optimized”, and “optimizing”. The system must ensure consistent keyword indexing and ranking. Develop a morphology-driven solution integrating inflectional and derivational analysis, finite-state morphological parsing for rule validation, and Porter Stemmer for normalization. Apply the approach to the given inputs and determine how the system achieves improved search relevance and reduced redundancy.
4. A large-scale information retrieval system processes input words: “compute”, “computer”, “computing”, and “computation” to enhance search accuracy.
Design a hybrid morphological pipeline integrating inflectional and derivational analysis, finite-state morphological parsing, and Porter Stemmer.
Decompose each word into root and affixes, represent transformations using state transitions, and apply stemming rules for normalization.
Ensure consistent root extraction across all variations for efficient indexing.
Determine the morphological breakdown, parsing transitions, and final normalized forms for each word.
5. A text mining system processes input words: “decide”, “decision”, “decisive”, and “deciding” to support semantic clustering and analysis.
Construct a morphology-based framework combining derivational rules, inflectional transformations, finite-state parsing, and Porter stemming.
Analyze how suffix variations affect meaning and grammatical structure while maintaining root consistency. Apply the approach to extract roots, validate transformations, and normalize the dataset. Derive the morphological structure, state transitions, and final stem for each input word.

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Task	20%	Clearly understands the given task/experiment without assistance	Understands task with minor clarification	Partial understanding; requires guidance	Unable to understand the task
Procedure & Execution	25%	Performs procedure accurately and independently with proper technique	Performs with minor errors	Requires guidance; makes procedural mistakes	Unable to execute the procedure
Observation & Result	20%	Records accurate observations and obtains correct results	Minor errors in observation or result	Incomplete or partially correct results	Incorrect or no results
Viva Voce / Conceptual Response	20%	Answers all questions confidently with strong conceptual clarity	Answers most questions with minor gaps	Limited responses; needs prompting	Unable to answer questions
Time Management & Presentation	15%	Completes task on time with neat and clear presentation	Completes with minor delay or presentation issues	Delayed or poorly presented work	Unable to complete on time

Q. No. / Criteria Level	Understanding of Task	Procedure & Execution	Observation & Result	Viva Voce / Conceptual Response	Time Management & Presentation	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Question 1: Morphological Normalization of "talked", "talking", "talks"

1.1 Given Words

The document processing system encounters three surface forms of the same lexeme: "talked", "talking", and "talks". All three are inflected forms of the base verb "talk".

1.2 Inflectional Morphology Analysis

Inflectional morphology modifies a word to express grammatical categories such as tense, aspect, number, or person, without changing its core meaning or part of speech. In this case:

- "talked" = talk + ed (past tense marker)
- "talking" = talk + ing (present participle / progressive aspect marker)
- "talks" = talk + s (third person singular present marker)

Since all three forms share the identical root "talk" and differ only in grammatical suffixes, they are treated as inflectional variants rather than derivational forms.

1.3 Porter Stemmer Steps Applied

Word	Porter Stemmer Rule Applied	Suffix Removed	Resulting Stem
talked	Step 1b: (*v*)ED → (delete ED, stem contains a vowel)	-ed	talk
talking	Step 1b: (*v*)ING → (delete ING, stem contains a vowel)	-ing	talk
talks	Step 1a: SSES/.../S → (delete plural/3rd-person S)	-s	talk

In each case, since the resulting stem "talk" already contains a vowel and does not end in a double consonant or a short CVC pattern, no further restoration (such as adding back an "e") is required.

1.4 Final Normalized Root

All three surface forms — "talked", "talking", "talks" — normalize to the single root form: talk

1.5 Role of Morphological Normalization in Information Retrieval

- Query–document matching: A user searching for "talk" will retrieve documents containing "talked", "talking", or "talks", improving recall.
- Index compression: Storing a single normalized token instead of multiple inflected variants reduces the size of the inverted index.
- Reduced redundancy: Term frequency statistics are consolidated under one root, giving more accurate relevance scoring (e.g., TF-IDF).
- Consistency: Normalization ensures that semantically identical occurrences are not treated as distinct terms during ranking and clustering.

Question 2: Hybrid Morphological Pipeline for "relational", "conditional", "happiness", "running"

2.1 Pipeline Design

The hybrid pipeline processes each word through four coordinated stages:

- Stage 1 – Derivational Rule Analysis: Identify affixes that change part of speech or create new lexical meaning (e.g., -al, -ness).
- Stage 2 – Inflectional Analysis: Identify grammatical suffixes (e.g., -ing) that do not change word class.
- Stage 3 – Finite-State Morphological Parsing: Model root + affix combinations as transitions between states in a finite-state automaton (FSA) to validate legal morpheme sequences.
- Stage 4 – Porter Stemmer: Apply rule-based suffix stripping to reduce each word to its final normalized stem.

2.2 Root and Affix Decomposition

Word	Morpheme Decomposition	Morphology Type
relational	relate + ion + al	Derivational
conditional	condition + al	Derivational
happiness	happy + ness	Derivational
running	run + ing	Inflectional

2.3 Finite-State Parsing Transitions

Each word is modelled as a path through an FSA with a Start state (S0), Root-recognition state (S1), Affix-attachment state(s) (S2, S3...), and an Accept state (SF):

- relational: S0 --relate--> S1 --ion--> S2 --al--> SF
- conditional: S0 --condition--> S1 --al--> SF
- happiness: S0 --happy(y→i)--> S1 --ness--> SF
- running: S0 --run(consonant doubling)--> S1 --ing--> SF

The FSA validates that each transition corresponds to a permissible morphotactic rule (e.g., that "-al" may attach after a nominal stem, or that consonant doubling is required before "-ing" in monosyllabic CVC roots), rejecting illegal affix sequences before stemming is applied.

2.4 Porter Stemmer Application

Word	Porter Rule	Final Stem
relational	Step 2: (m>0) ATIONAL → ATE	relate
conditional	Step 2: (m>0) TIONAL → TION	condition
happiness	Step 1c: (*v*) Y → I ; Step 3: (m>0) NESS → (delete)	happi
running	Step 1b: doubled consonant + ING → single consonant	run

2.5 Contribution of Each Technique

- Derivational rules capture meaning/word-class change (e.g., condition → conditional as noun→adjective) that pure inflection would miss.
- Inflectional analysis correctly isolates grammatical suffixes (e.g., -ing in "running") without altering the lexical category of the root.
- Finite-state parsing enforces structural legality of morpheme ordering, preventing over- or under-stemming caused by blind suffix stripping.
- The Porter Stemmer provides a fast, rule-based mechanical reduction to a usable indexing stem once the morphological structure has been validated.
- The hybrid morphological pipeline combines linguistic knowledge with computational efficiency.
- Derivational analysis captures semantic changes, while inflectional analysis identifies grammatical variations without affecting the original meaning.
- Finite-State Morphological Parsing validates whether the sequence of morphemes follows valid language rules, thereby preventing incorrect segmentation.
- Finally, the Porter Stemmer provides a fast and efficient normalization process suitable for large datasets.
- By integrating all four techniques, the system achieves higher accuracy in root extraction, reduces indexing errors, improves search performance, and supports advanced NLP tasks such as machine translation, text mining, document clustering, and information retrieval.

Question 3: SEO Morphology-Driven Solution for "optimize", "optimization", "optimized", "optimizing"

3.1 Problem Context

An enterprise SEO indexing system must treat "optimize", "optimization", "optimized", and "optimizing" as a single semantic keyword cluster so that search ranking is not fragmented across near-identical query terms.

3.2 Inflectional and Derivational Analysis

Word	Decomposition	Morphology Type
optimize	optim + ize	Base / Derivational suffix
optimization	optim + ize + ation	Derivational (verb → noun)
optimized	optim + ize + ed	Inflectional (past tense)
optimizing	optim + ize + ing	Inflectional (progressive)

3.3 Finite-State Morphological Parsing (Rule Validation)

- optimize: S0 --optim--> S1 --ize--> SF
- optimization: S0 --optim--> S1 --ize--> S2 --ation--> SF
- optimized: S0 --optim--> S1 --ize--> S2 --ed--> SF
- optimizing: S0 --optim--> S1 --ize--> S2 --ing--> SF

The FSA confirms that "-ation", "-ed", and "-ing" are all valid continuations only after the verb-forming suffix "-ize" has been accepted, ensuring the parser does not mistakenly split the root at an invalid boundary (e.g., "optimiz-ation" instead of "optim-iz-ation").

3.4 Porter Stemmer Normalization

Word	Porter Rule Applied	Normalized Stem
optimize	No change (already minimal form)	optimize
optimization	Step 2: (m>0) IZATION → IZE	optimize
optimized	Step 1b: (*v*) ED → delete; restore E after Z	optimize
optimizing	Step 1b: (*v*) ING → delete; restore E after Z	optimize

3.5 Improved Search Relevance and Reduced Redundancy

- All four variants collapse to the single index token "optimize", so a search query using any inflected form retrieves the full set of relevant pages.
- Keyword frequency for ranking (TF-IDF / BM25) is computed on the consolidated stem, avoiding artificial dilution of relevance scores across surface variants.
- Index size and storage cost are reduced because duplicate near-identical tokens are not stored separately.
- Search relevance improves because content using different grammatical forms of the same keyword is ranked coherently rather than being fragmented.
- From an SEO perspective, morphology-based normalization ensures that webpages containing different forms of the same keyword are treated as equally relevant.
- This prevents search engines from splitting ranking signals across multiple word variants and instead consolidates them into a single keyword representation.
- As a result, webpages achieve better visibility in search results, users receive more accurate responses, and duplicate indexing is minimized.
- Additionally, the reduced vocabulary size lowers storage requirements and speeds up query processing, making the overall search engine more scalable and efficient for handling millions of indexed web pages.

Question 4: Hybrid Pipeline for "compute", "computer", "computing", "computation"

4.1 Morphological Breakdown

Word	Decomposition	Morphology Type
compute	comput + e	Base verb form
computer	comput + er	Derivational (verb → agent noun)
computing	comput + ing	Inflectional (progressive)
computation	comput + ation	Derivational (verb → abstract noun)

4.2 Finite-State Parsing Transitions

- compute: S0 --comput--> S1 --e--> SF
- computer: S0 --comput--> S1 --er--> SF
- computing: S0 --comput--> S1 --ing--> SF
- computation: S0 --comput--> S1 --ation--> SF

The shared state S1 (reached via the common root "comput") demonstrates that all four surface forms diverge only at the final affix-attachment transition, confirming a single underlying lexeme for indexing purposes.

4.3 Porter Stemmer Application

Word	Porter Rule Applied	Final Stem
compute	Step 5a: (m>1) E → delete	comput
computer	No standard suffix rule for agentive -er; treated as root + er	comput
computing	Step 1b: (*v*) ING → delete	comput
computation	Step 4: (m>1) ION → delete (preceded by T)	comput

4.4 Consistent Root Extraction and Indexing Benefit

- All four words normalize to the common stem "comput", allowing the retrieval system to index them under one term.
- Finite-state validation prevents incorrect segmentation (e.g., ensures "computation" is not split as "comp-utation").
- Combining derivational awareness (agent noun "computer", abstract noun "computation") with mechanical stemming preserves interpretability while still achieving indexing consistency.
- The consolidated stem improves recall for queries such as "computing systems" or "computer science" by matching all morphological variants.
- Consistent root extraction is essential in large-scale information retrieval systems because users often search using different grammatical forms of the same concept.
- By mapping all variants such as **compute**, **computer**, **computing**, and **computation** to a common stem, the indexing process becomes more compact and efficient.
- This improves document retrieval accuracy, reduces duplicate entries in the index, and enhances ranking algorithms like TF-IDF and BM25.
- Moreover, the hybrid pipeline ensures that meaningful derivational relationships are preserved while maintaining a unified representation of related words, thereby supporting semantic search and knowledge discovery applications.

Question 5: Morphology Framework for "decide", "decision", "decisive", "deciding"

5.1 Morphological Structure

Word	Decomposition	Morphology Type	Grammatical Role
decide	decide (base)	Root verb	Verb
decision	decide + sion	Derivational	Noun
decisive	decide + sive	Derivational	Adjective
deciding	decide + ing	Inflectional	Verb (progressive)

5.2 Effect of Suffix Variation on Meaning and Grammar

- "-sion" converts the verb "decide" into the abstract noun "decision" (the act/result of deciding).
- "-sive" converts the verb into the adjective "decisive" (characterized by firm decision-making).
- "-ing" is purely inflectional, marking progressive aspect while keeping "deciding" as a verb form.

Although all four words are conceptually related to the semantic root "decide", the derivational suffixes change part of speech and add nuanced meaning, while the inflectional suffix only changes grammatical aspect.

5.3 Finite-State Parsing Transitions

- decide: S0 --decide--> SF
- decision: S0 --decid--> S1 --sion--> SF
- decisive: S0 --decis--> S1 --ive--> SF
- deciding: S0 --decid--> S1 --ing--> SF

The FSA validates that "-sion" and "-ive" attach to the allomorph "decis-", while "-ing" attaches to "decid-", capturing the spelling-change (d/s alternation) that a purely mechanical stemmer may not distinguish.

5.4 Porter Stemmer Application and Normalization

Word	Porter Rule Applied	Resulting Stem
decide	Step 5a: (m>1) E → delete	decid
decision	Step 4: (m>1) ION → delete (preceded by S)	decis

Word	Porter Rule Applied	Resulting Stem
decisive	Step 4: (m>1) IVE → delete	decis
deciding	Step 1b: (*v*) ING → delete	decid

5.5 Analysis and Root Consistency

- The Porter Stemmer alone produces two distinct stems — "decid" and "decis" — because it operates mechanically on surface spelling and cannot recognize that both derive from one lexeme.
- The hybrid framework resolves this by using derivational/inflectional analysis and finite-state validation to map both stem variants back to the single canonical lemma "decide" during semantic clustering.
- This demonstrates why combining rule-based derivational knowledge with mechanical stemming yields more reliable root consistency than Porter stemming in isolation, which is critical for accurate semantic clustering in text mining.
- The proposed hybrid framework demonstrates that relying solely on mechanical stemming is often insufficient for complex morphological variations.
- By incorporating derivational analysis, inflectional analysis, and finite-state parsing, the system successfully recognizes that **decide**, **decision**, **decisive**, and **deciding** belong to the same lexical family despite spelling differences.
- This improves semantic clustering, document classification, and topic modeling by grouping related words under a common concept.
- Consequently, the framework enhances NLP applications such as search engines, question answering systems, text summarization, and intelligent recommendation systems by providing more meaningful and context-aware language representations.

